

Seong Lee

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RESEARCH INTERESTS

Transport phenomena and rheology of complex fluids

EDUCATION

Seoul National University (SNU), Seoul, Korea

Mar. 2020 – Feb. 2027

B.S. in Chemical and Biological Engineering, *Summa Cum Laude*

(expected)

- GPA: 4.04/4.30 (overall) | 4.17/4.30 (major)
- Graduate courses: Transport Phenomena, Polymer Rheology, Polymeric Materials

PUBLICATIONS

† Equal contribution; * Corresponding author

- [J1] C. Hyun[†], **S. Lee**[†], W. Jo[†], and J. Nam^{*}. Understanding polyvinylpyrrolidone dispersion mechanism in $\text{LiMn}_x\text{Fe}_{1-x}\text{PO}_4$ slurry for electrode processing and performance. Manuscript in preparation.
- [P1] J. Shin, **S. Lee**, et al. Method for producing alginic acid extract derived from seaweed. KR Patent 10-2969478, issued May 21, 2026. [doi:10.8080/1020250161063](https://doi.org/10.8080/1020250161063)
- [P2] J. Shin, **S. Lee**, et al. Method for improving the purity of alginic acid through separation and purification of impurities. KR Patent 10-2953408, issued Apr. 13, 2026. [doi:10.8080/1020250161058](https://doi.org/10.8080/1020250161058)

CONFERENCE PRESENTATIONS

- [C1] **S. Lee**, C. Hyun, W. Jo, and J. Nam. Influence of PVP molecular weight and concentration on LMFP slurry dispersion and electrode performance. *Asian Coating Workshop*, Kaohsiung, Taiwan, May 13-15, 2026. **1st Place, Best Presentation Award.**
- [C2] **S. Lee**, C. Hyun, W. Jo, and J. Nam. Effect of PVP molecular weight and concentration on the properties of LMFP slurries and electrodes. *KIChE Spring Meeting*, Jeju, Korea, Apr. 22-24, 2026.

RESEARCH & WORK EXPERIENCE

Microfluidics and Coating Process Laboratory, SNU, Korea

Undergraduate Researcher, *Advisor: Prof. Jaewook Nam*

Aug. 2025 – Present

- Topic: **Slurry rheology** and **electrochemical performance** of olivine-based LIB cathodes
 - Identified steric stabilization as the dispersion mechanism of PVP in LMFP slurries, attributing MW-dependent dispersion efficiency to polymer self-association. [C2]
 - Examined degradation pathways via cell-level electrochemical impedance spectroscopy. [C1] [J1]
 - Investigating blends of carboxyl-modified and homopolymer PVDF to decouple binder roles for bilayer electrode design.
 - Quantified trade-offs and optimal balance in carbon-black/CNT hybrid conductive networks.

Supercritical Fluid Process Laboratory, SNU, Korea

Undergraduate Researcher, *Advisor: Prof. Emeritus Youn-Woo Lee*

Dec. 2024 – Jan. 2026

- Topic: **Reaction kinetics** and **separation processes**
 - Developed a [web-based Hunter-Nash simulator](#) for liquid-liquid extraction education, automating equilibrium fitting, graphical stage stepping, and sensitivity analysis of operating conditions.
 - Characterized coated-hanji degradation kinetics via accelerated-aging colorimetry, showing beeswax deacidification raises the activation energy, in support of *Annals of the Joseon Dynasty* conservation.

- Determined intrinsic kinetics for the oxidation and cyclization steps of oxfendazole synthesis, enabling Levenspiel-based PFR designs for commercial-scale (20 MT/yr) production with Kolon Life Science.

Plantner Co., Ltd. (Seaweed-derived Agri-tech biomaterials startup), Seoul, Korea

Researcher, R&D Team

Apr. 2025 – Oct. 2025

- Developed formaldehyde-free, room-temperature alginate extraction process, achieving 42% yield. [P1]
- Designed dual-polarity solvent system for selective pigment removal, improving extract purity. [P2]
- Drafted R&D grant proposals and investor materials for product scale-up and international outreach, contributing to the company's **2026 Edison Awards (Silver)**.

SELECTED AWARDS & HONORS

Undergraduate Research Grant, SNU

May 2026

Awarded \$3,000 through the Student-Directed Research Program.

The Education and Research Foundation Scholarship, College of Engineering, SNU

2025, 2026

Awarded \$2,400 (\$1,200 × 2 years); sole nominee from the department.

Merit-based Scholarship, SNU

2024, 2025

Full-tuition award based on academic ranking.

Excellence Award, Next-Generation Engineer, IPESK

Oct. 2025

Awarded for academic excellence; top 2 of 20 nominees from the department.

Gold-collar Engineer Award, Korean Society for Engineering Education

Sep. 2025

One of 9 nationwide recipients for interdisciplinary engineering excellence.

Excellence Award, Undergraduate Research Program Utilizing Advanced Equipment, SNU

Jan. 2022

Quantified humidity-induced PbI₂ formation in MAPbX₃ perovskites via PXRD and SEM/EDS.

ACTIVITIES & SOCIETIES

Peer Tutoring Program, SNU

2022, 2025

Provided 100+ hours of weekly academic support; awarded **Outstanding Tutor**.

- Organic Chemistry (Fall 2025), Process Fluid Mechanics (Spring 2025), Basic Chemistry (Spring 2022)

STEM: SNU Tomorrow's Engineers Membership, SNU

Sep. 2024 – Present

Honorary Member, Voluntary Service Division

- Led two annual career mentoring events (*CBEyond: the Day*) with the student council, featuring alumni speakers across industry, academia, and law, hosting 60+ participants (4.7/5 satisfaction).
- Mentored 80+ sophomores/juniors (3 events) and 50+ local high school students (2 events); managed the \$5,000 *STEM Vision Scholarship* for 5 selected recipients.

Student Volunteer, *International Symposium on Applied Rheology*, Seoul, Korea

May 7, 2026

Korea Leader Mentoring Network, Korea Student Aid Foundation

Apr. 2025 – Nov. 2025

Air Intelligence Operator, Republic of Korea Air Force

Sep. 2022 – Jun. 2024

Managed intelligence data and supported unit operations; honorably discharged as Sergeant.

SKILLS

Programming: Python, MATLAB, C++, LaTeX, HTML/CSS

Modeling & Simulation: Aspen Plus, Aspen HYSYS, PRO/II, COMSOL

Characterization: Rheometry, DLS, zeta potential, galvanostatic cycling, EIS, SEM/EDS, PXRD

Languages: English (Fluent; TOEFL 110), Korean (Native), Japanese (Fluent)